Problem 3

Problem 3

Problem 1 *Given that:*

$$\lim_{x\to-\infty} f(x) = 0 \qquad \lim_{x\to-3^-} f(x) = \infty \qquad \lim_{x\to-3^+} f(x) = -\infty \qquad \lim_{x\to4^-} f(x) = \infty$$

$$\lim_{x\to4^+} f(x) = -\infty \qquad f(1) = 1 \qquad f(5) = -2 \qquad \lim_{x\to9} f(x) = 3$$

$$f'(1) \neq 0 \qquad f'(7) = 0 \qquad f'(x) = 2 \text{ for } x > 9 \qquad f''(1) = 0$$

the domain of f is: $(-\infty,-3)\cup(-3,4)\cup(4,9)\cup(9,\infty)$ and f is continuous on its domain. The following sign chart for the first and second derivatives of f:

find the following:

(a) *Critical points.*

(b) *Intervals where f is increasing/decreasing.*

(c) *Local extrema.*

(d) *Inflection points.*

(e) *Intervals of concavity.*

(f) *Sketch the graph of f.*